Sources and Resources Documentation

Project: A Day with ElliQ: Inside an Elder's Life with an AI Companion

Course: Interactive News Project

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Major:HCI

Live Project URL: https://folk-fig-91932575.figma.site

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1. PRIMARY NEWS SOURCES

AP News Articles (Primary Source Material)

All narrative content and quotes are based on two AP News reports:

1. **"Companion robots are trying to combat loneliness, but do they offer real support or just an illusion?"**
    - Publisher: Associated Press
    - Date: October 2024
    - Access: AP News official website
    - Usage: Primary source for all four elder profiles (Joyce Loaiza, Deanna Dezern, Marie Broadbent, Jan Worrell), ElliQ feature descriptions, and direct quotes
2. **"AI companion robots combat loneliness among elderly"**
    - Publisher: Associated Press
    - Date: 2024
    - Access: AP News official website
    - Usage: Background information on ElliQ technology, New York State pilot program details, and expert perspectives

Citation Format: All scenes include proper AP News attribution in accordance with journalism standards and academic requirements.

2. THEORETICAL FRAMEWORK

Academic Theory

- **Erikson's Psychosocial Development Theory**
  - Stage: "Ego Integrity vs. Despair" (late adulthood)
  - Application: Ethical reflection scene analyzing AI companionship through developmental psychology lens
  - Usage: Framework for understanding the emotional and psychological impact of AI companions on elderly users

3. IMAGES AND VISUAL ASSETS

Imported from Figma Design

All character photos and ambient images were imported from original Figma design files:

- Joyce Loaiza portrait (81 years old)
- Deanna Dezern portrait (83 years old)
- Marie Broadbent portrait (92 years old)
- Jan Worrell portrait (83 years old)
- Morning ambient photos (breakfast scenes)
- Afternoon ambient photos (park reading scenes)
- Evening ambient photos (sunset/twilight scenes)

Source Attribution: Photos are properly credited in the "Four Lives Changed" scene with photo credits displayed below each portrait.

Unsplash Stock Images

Used under Unsplash License for supplementary visual elements:

- Museum tour images (The Louvre, MoMA)
- Book cover representations
- Background ambience photos

Implementation: Images loaded via unsplash_tool API integration

4. AUDIO AND SOUND

Text-to-Speech Technology

- **Technology:** Web Speech API (browser-native)

- **Implementation:** Custom textToSpeech.ts utility
- **Voice Characteristics:**
    - Preferably female US English voice
    - Rate: 0.95 (slightly slower for clarity)
    - Pitch: 1.1 (slightly higher for friendliness)
    - Volume: 0.9

Audio Content Scenes

1. **Morning Scene - Joyce's Mood Check**
    - ElliQ greeting and conversation
    - Joke delivery
2. **Afternoon Scene - Marie's Experience (Full Audio)**
    - 5-part sequential narration
    - Describes visual world for blind user
    - Demonstrates accessibility features
3. **Evening Scene - Video Call Simulation**
    - Emotional support dialogue

Note: All audio is generated in real-time using browser TTS; no pre-recorded audio files used.

5. TECHNICAL FRAMEWORKS AND LIBRARIES

Core Technologies

- **React 18+** - UI framework
- **TypeScript** - Type-safe development
- **Vite** - Build tool and dev server

UI Component Library

- **shadcn/ui** - Used under MIT License
    - Button, Card, Dialog, Tabs components
    - Accordion, Progress, Slider components

Animation and Interaction

- **Motion (Framer Motion)** - Animation library

    - Package: motion/react
    - Usage: Scene transitions, scroll animations, interactive elements
- **Lucide React** - Icon library
    - Icons: Coffee, Heart, Eye, BookOpen, Music, Video, Sparkles, Volume2, etc.

Data Visualization

- **Recharts** - Chart library
    - Usage: Data visualization in Scene 8 (Data & Impact)
    - Charts: User satisfaction metrics, loneliness reduction statistics

Styling

- **Tailwind CSS v4.0** - Utility-first CSS framework
- **Custom CSS** - Typography system in /styles/globals.css

6. INTERACTIVE ELEMENTS AND DATA

Mock Data and Simulations

All interactive elements use realistic mock data:

1. **Medication Reminder System**
    - Mock medication schedule
    - Interactive reminder responses
2. **Music Therapy Options**
    - Classical Piano, Jazz, Nature Sounds, 1960s Oldies
    - Simulated playback interface
3. **Poetry Reading**
    - Emily Dickinson - "Hope is the thing with feathers"
    - Robert Frost - "The Road Not Taken"
4. **Virtual Museum Tours**
    - The Louvre (Paris)
    - MoMA (New York)
5. **Statistical Data (Scene 8)**
    - Based on AP News reporting of ElliQ program results
    - User satisfaction metrics

    - Loneliness reduction statistics

7. CODE STRUCTURE AND COMPONENTS

Custom Components Created

- **Scene1IntroMinimal.tsx** - Opening scene
- **SceneElderStoriesRich.tsx** - Four lives profiles
- **SceneElliQ3DEnhanced.tsx** - ElliQ robot introduction
- **SceneContextBackground.tsx** - Background context
- **SceneJoyceMoodInteraction.tsx** - Joyce's morning
- **SceneMorningWithElliQ.tsx** - Morning routine
- **SceneAfternoonWithElliQ.tsx** - Afternoon activities (with Marie's audio experience)
- **SceneEveningWithElliQ.tsx** - Evening reflection
- **SceneEthicsEnglish.tsx** - Ethical considerations
- **Scene8Data.tsx** - Data visualization
- **Scene9Ending.tsx** - Conclusion
- **Scene10Credits.tsx** - Credits and references
- **NavigationMenu.tsx** - Scene navigation
- **DynamicBackground.tsx** - Animated backgrounds

Utility Files

- **utils/textToSpeech.ts** - ElliQ voice synthesis
    - Custom TTS service wrapper
    - ElliQ-specific voice presets
    - Error handling for speech API

8. RESEARCH AND REFERENCES

Loneliness and Elderly Care Research

- **Context:** AP News articles cite various studies on elderly loneliness epidemic
- **Statistics:** 800 ElliQ units distributed in New York State pilot program
- **Outcomes:** User testimonials and program results from AP reporting

AI Ethics Considerations

- **Privacy concerns** - Data collection and voice recording
- **Dependency risks** - Potential over-reliance on AI vs. human connection
- **Digital divide** - Access equity questions
- **Emotional authenticity** - Real connection vs. simulated companionship

Analysis Framework: Balanced exploration presented through interactive quiz and reflection prompts

9. DESIGN PRINCIPLES

Visual Design

Color Palette:

- Morning: Amber/Orange gradients (sunrise warmth)
- Afternoon: Blue/Cyan gradients (daytime clarity)
- Evening: Purple/Pink gradients (sunset serenity)
- Night: Deep purple/indigo (nighttime calm)

Typography: Custom type scale defined in globals.css

Layout: Responsive design with mobile-first approach

User Experience

- **Navigation:** Smooth scroll-based progression with anchor navigation
- **Interactivity:** Multiple interaction types (buttons, choices, audio, data viz)
- **Accessibility:** Semantic HTML, keyboard navigation, screen reader considerations

10. PROJECT DEPLOYMENT

Hosting Platform

- Figma Make deployment platform
- Automatic build and deployment from codebase

11. DEVELOPMENT WORKFLOW

Version Control

- Iterative development with scene-by-scene implementation

- Regular testing of TTS functionality
- Cross-browser compatibility testing (Chrome, Firefox, Safari, Edge)

Quality Assurance

- Audio playback testing across different browsers
- Responsive design testing on multiple screen sizes
- Accessibility audit for keyboard and screen reader navigation
- Citation accuracy verification against source AP News articles

12. ACADEMIC INTEGRITY STATEMENT

This project:

- ✅ Properly attributes all AP News source material
- ✅ Uses only licensed or open-source technologies
- ✅ Includes photo credits for all imported images
- ✅ Generates audio dynamically (no copyrighted audio files)
- ✅ Creates original interactive experiences based on factual reporting
- ✅ Maintains journalistic integrity with accurate quotes and context

All content is used for educational purposes as part of an interactive news assignment.